

Introduction to Computational Thinking and Python Programming

Instructor: Sarom Leang, PhD (Teacher/Professor)

Specialist: Eesha Randhawa

Innovation Hall 203



Pronouns

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- Specialist: Eesha Randhawa



Schedule

- 10:00 AM – 10:15 AM (Homeroom)
- 10:15 AM – 11:35 AM (G1)
- 11:40 AM – 12:35 PM (Lunch)
- 12:40 PM – 02:00 PM (G2)



Student Expectations

- **NO FOOD**
- **NO DRINKS** (on the table)
- Be respectful to individuals and property
- Be open to learning
- Be open to not understanding
- Be patient with yourself
- Ask questions
- Explore
- **Embrace failure**



Course Overview (cont.)

Exploratory Topics

- ~~Session #1 – November 2, 2024~~
 - ~~Entrance Survey~~
 - ~~How Computers Work~~
- ~~Session #2 – December 7, 2024~~
 - ~~Parallel Computing~~
- ~~Session #3 – February 1, 2025~~
 - ~~Generative Artificial Intelligence~~
- Session #4 – March 22, 2025
 - Quantum Computing
- Session #5 – April 12, 2025
 - Future of Computing
 - Exit survey

Topics

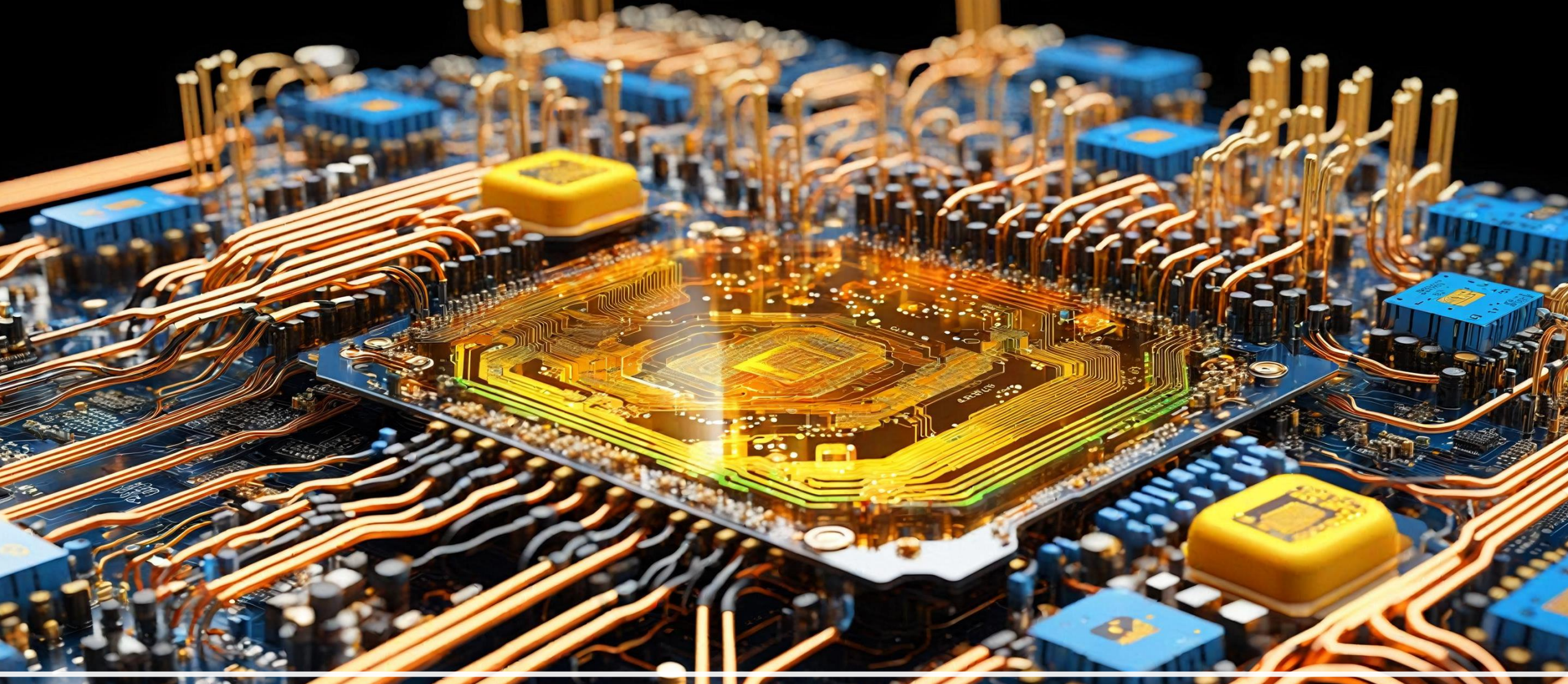
- Algorithms and Problem Solving
- Debugging and Troubleshooting
- Loops
- Algorithms and Syntax
- Variables and Conditionals
- Variable Arithmetic
- Conditionals (If/Else)
- Compound Conditionals



Recap: Session 3

- Generative AI
 - A type of artificial intelligence that can create new content that is similar to the original data it was trained on
 - Learn patterns and relationships in data, and uses that knowledge to generate new content
 - Can produce content that is incorrect or incoherent (hallucinate)
 - Generate responses based on statistical patterns rather than comprehension
- Data (raw data) -> Information (processed) -> Knowledge (insight)
- Understand the rules governing generative AI use in your classes/schools/work place





Quantum Computing

Quantum Computing

Classical Computing

- Data is stored as bits (0 or 1)
- Data is processed **sequentially**
- Processing power increases **linearly** with the number of bits (32-bit processor, 64-bit processor,

Quantum Computing

- Data is stored as qubits (**superposition** of 0 and 1)
- Data can be processed in **parallel** (**superposition** and **entanglement**)
- Processing power increases **exponentially** with the number of qubits



Superposition

- Describes a state where a quantum system (e.g., qubits) **exists** in a **combination of multiple states simultaneously**
 - Classical bit: a coin that can be either heads **OR** tails
 - Qubit: a special coin that can be both heads **AND** tails at the same time



Entanglement

- Is a special **connection between two quantum particles** that lets them “talk” to each other **instantly**, no matter how far apart they are
 - Classical communication: sending a message from New York to Los Angeles via email or phone
 - Quantum entanglement: sending a message from New York to Los Angeles instantly, without using any physical medium



Superposition and Entanglement

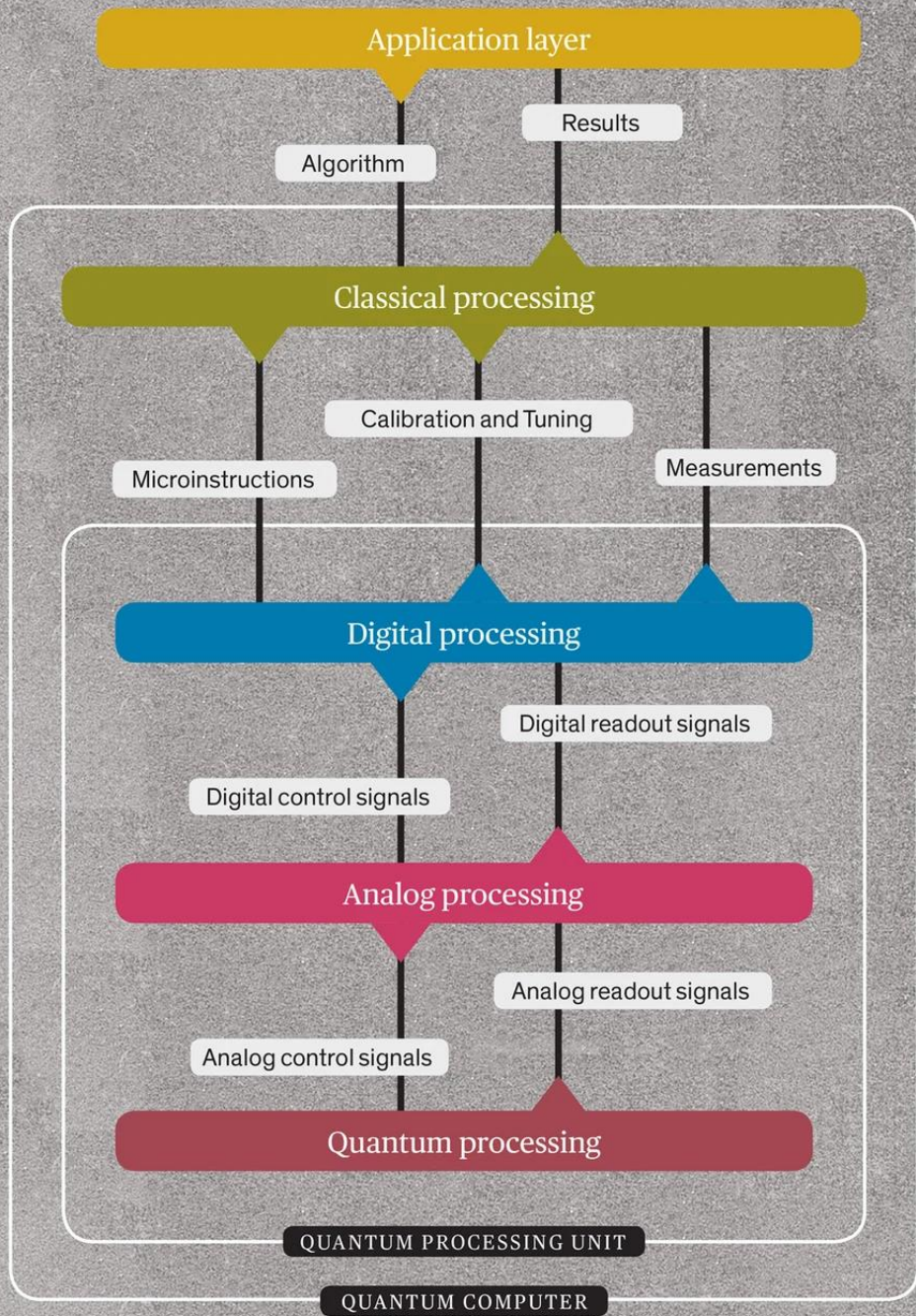
- Superposition allows for **processing multiple possibilities simultaneously**
 - Combination lock with 10 possible combinations
 - Classical – try each combination one at a time (sequential/serial)
 - Quantum – could try all 10 combinations simultaneously (parallel)
- Entanglement enables **quantum parallelism**
 - Two entangled qubits you can perform operations on both qubits simultaneously, effectively processing four possibilities at once: (00, 01, 10, and 11)
 - With three entangled qubits, eight possibilities at once: (000, 001, 010, 011, 100, 101, 110, 111)



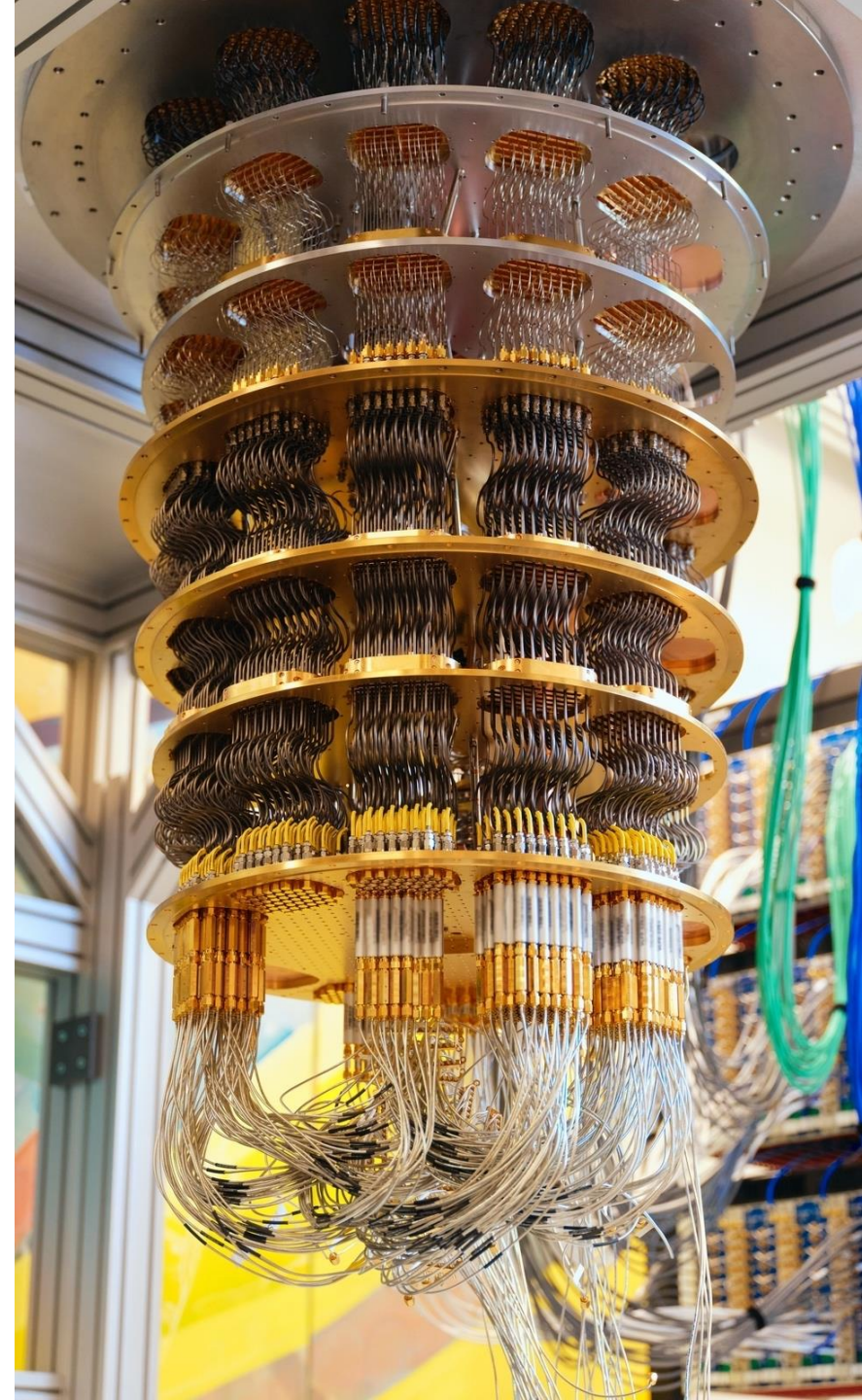
Power of Quantum Computing

- Exponential scaling
 - As the number of qubits increases, the power of quantum computing grows exponentially – each qubit doubles the number of possible states that can be processed simultaneously
 - 1 qubit: 2 possible states (0 and 1)
 - 2 qubits: 4 possible states (00, 01, 10, and 11)
 - 3 qubits: 8 possible states (000, 001, 010, 011, 100, 101, 110, and 111)
 - n qubits: 2^n possible states
- Ideal applications
 - Large dataset – process massive amounts of data simultaneously (ML or DA)
 - Complex simulations – simulate complex systems
 - Cryptography – create unbreakable quantum encryption methods





Quantum Computer



Today

- Log into Ozaria
- Complete Chapters 1-4

